

Heterogeneous Difference-in-Differences in Stata

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Stata

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Introduction

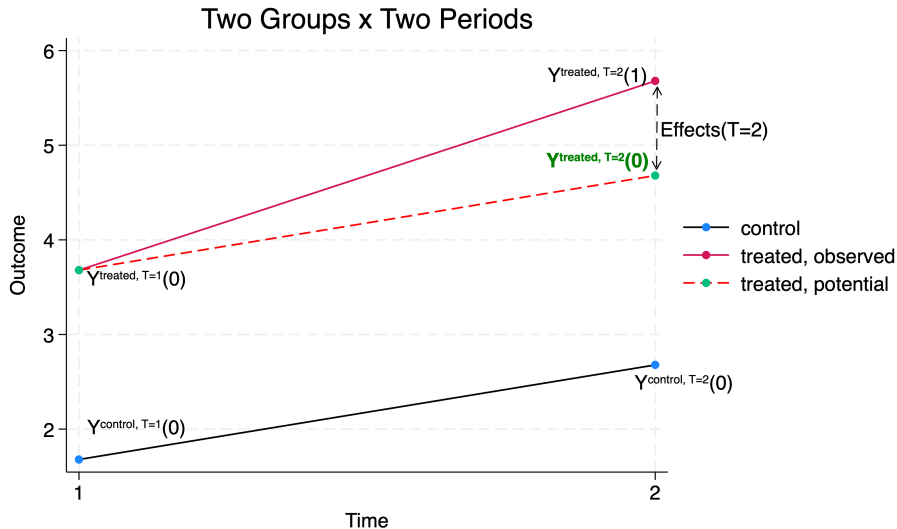
Setup

- Estimate treatment effects using panel data or repeated cross-section
- Treatments may start at different times
- Staggered treatment (once treated, always treated)

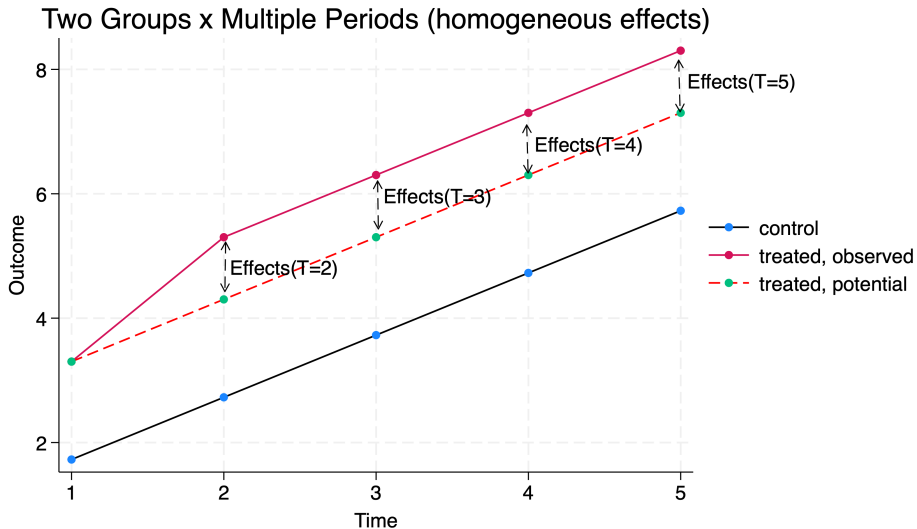
Group/Cohort: units in the same group start the treatment at the same time, different groups start treatments at different times

Objective: Estimate treatment effects for the treated groups

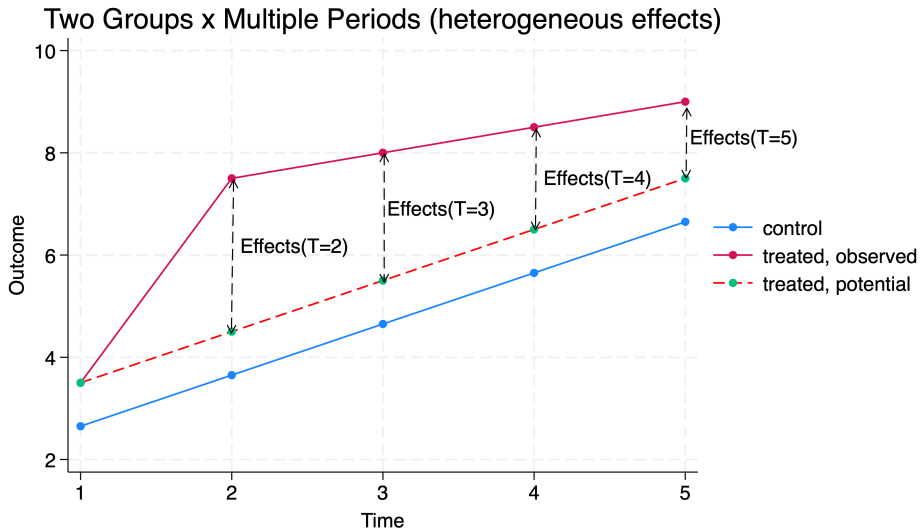
Canonical DID



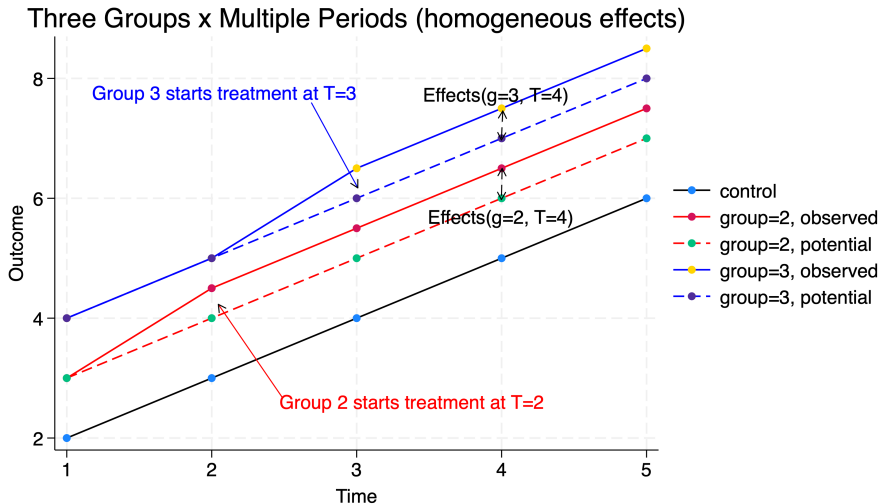
Homogeneous effects across time



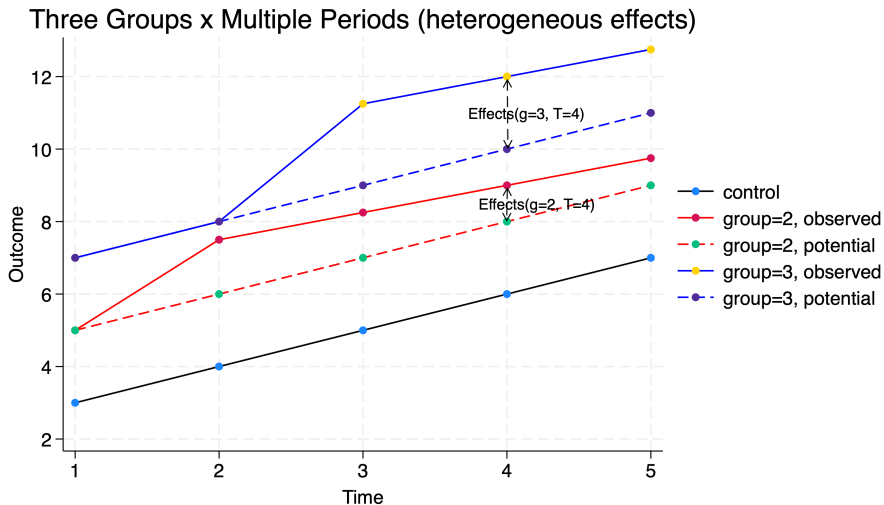
Heterogeneous effects across time



Homogeneous effects across groups and time



Heterogeneous effects across groups and time



Problematic TWFE

$$y_{it} = \theta_t + \eta_i + d_{it}\alpha + v_{it}$$

But,

$$\alpha = \sum w_k \text{Good_DID}_k + \sum w_j \text{Bad_DID}_j$$

- ① Newly treated relative to the control group (good)
- ② Newly treated relative to the not-yet treated group (good)
- ③ Newly treated relative to already treated group (bad)

When is TWFE good?

- 1 There are **only two periods**
- 2 The treatment effects are **homogenous across both groups and time**

Overview of heterogeneous DID in Stata 18

Estimation:

- 1 `xthdidregress` and `hdidregress` for panel data and repeated cross-section data
- 2 Four estimators: `ra`, `ipw`, `aipw` in Callaway and Sant'Anna (2021) and `twfe` in Wooldridge (2021)

Post-estimation:

- 1 `estat atetplot`: visualize ATETs
- 2 `estat aggregation`: aggregate ATETs along different dimensions
- 3 `estat ptrends`: pre-treatment parallel trend tests
- 4 `estat sci`: simultaneous CI for RA, IPW, and AIPW estimators

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Increasing minimum wage and young employment

- **Outcome:** county-level employment for young workers
- **Treatment:** minimum wage restrictions introduced by State government; see Callaway and Sant'Anna (2021)
- **Multiple periods:** 2002 - 2007 (6 years)
- **Multiple treatment timings:** 2004, 2006, 2007

```
xthdidregress aipw
```

Define covariates

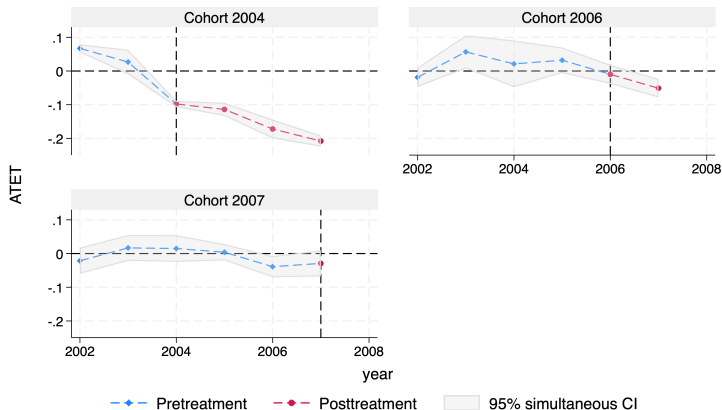
```
global covars i.region pop medinc white hs pov ///  
              c.pop#c.pop c.medinc#c.medinc
```

Use AIPW estimator

```
xthdidregress aipw (lemp $covars) (treat $covars), group(state)
```

- Adding covariates for conditional parallel trend
- There are 18 ATET(g,t)'s ($6 \text{ years} \times 3 \text{ cohorts}$)
- Standard errors are adjusted by clusters of state

estat atetplot, sci



- Specify option `sci` for simultaneous confidence intervals
- For cohorts 2004 and 2006, minimum wage restriction decreases the employment rate for young workers

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Summarize ATET(g,t)'s

- How do the ATETs vary with the length of **exposure to the treatment**? (event study)
- How do the ATETs vary with **cohorts**? (does start treatment earlier matter?)
- How do the ATETs vary with **time**? (Good year vs. lousy year)
- Overall ATETs across time and cohorts

We can express the aggregations as a weighted mean of all ATETs

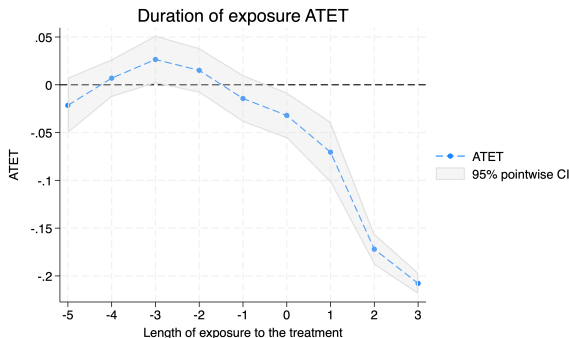
$$\theta = \sum_{g \in \mathbf{G}} \sum_{t=2}^T \underbrace{w(g, t)}_{\text{weight}} ATET(g, t)$$

Event study

- Let $e = t - g$ be the length of exposure to the treatment.

$$\theta(\mathbf{e}) = \sum_{g \in \mathbf{G}} \underbrace{\mathbb{1}(g + e \leq T) \Pr(G = g | G + e \leq T)}_{\text{proportions used to estimate ATET}(g, g+e)} \text{ATET}(g, g + e)$$

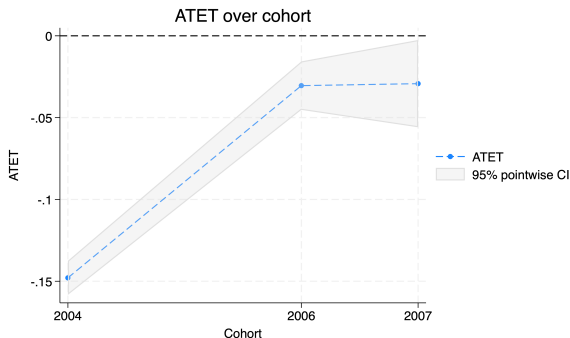
- estat aggregation, weights(cohort) **dynamic** graph



ATETs over cohort

$$\theta(\mathbf{g}) = \sum_{t=g}^T \underbrace{\frac{1}{T - g + 1} \mathbb{1}(t \geq g)}_{\text{proportions used to estimate post-treatment ATET}(g,t)} ATET(g, t)$$

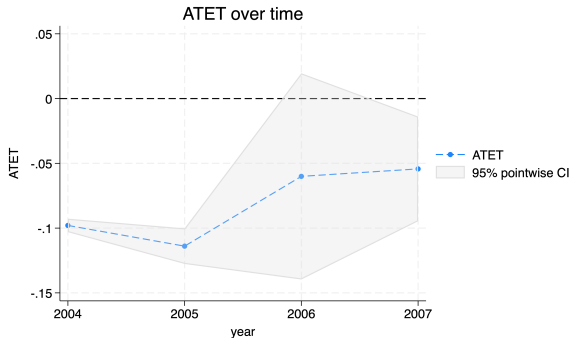
- estat aggregation, weights(cohort) **cohort** graph



ATETs across time

$$\theta(\mathbf{t}) = \sum_{g \in \mathbf{G}} \mathbb{1}(t \geq g) \Pr(G = g | G \leq t) ATET(g, t)$$

- estat aggregation, weights(cohort) **time** graph



Overall aggregations

- A single number to summarize ATET's

$$\theta = \frac{1}{\kappa} \sum_{g \in \mathbf{G}} \sum_{t=2}^T \mathbb{1}(t \geq g) \Pr(G = g | G \leq T) ATET(g, t)$$

- `estat aggregation, weights(cohort)` **overall**

```
. estat aggreg, overall weights(cohort)
```

```
Overall ATET
```

```
Number of obs = 15,988
```

```
(Std. err. adjusted for 29 clusters in state)
```

lemp	ATET	Robust std. err.	z	P> z	[95% conf. interval]	
treat (1 vs 0)	-.062811	.0256879	-2.45	0.014	-.1131582	-.0124637

Note: Aggregation weights vary across cohorts.

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Potential outcome framework

Some notations:

- $y_{i,t}(g)$ is unit i 's potential outcome at time t if it starts treatment at time g
- $y_{i,t}(\infty)$ is unit i 's potential outcome at time t if it is never treated
- G_i indicates unit i 's cohort (when the treatment starts), and it is one element in $\mathbf{G} = \{2, \dots, G, \infty\}$
- $y_{i,t}$ is unit i 's observed outcome at time t

$$y_{i,t} = \underbrace{\mathbb{1}(t < G_i)y_{i,t}(\infty)}_{\text{before treatment}} + \underbrace{\mathbb{1}(t \geq G_i)y_{i,t}(G_i)}_{\text{after treatment}}$$

Heterogeneous ATETs

$$ATET(g, t) = \mathbf{E}[y_{i,t}(g) - y_{i,t}(\infty) | G_i = g]$$

Remarks

- $ATET(g, t)$ is a function of two arguments: cohort g and time t
- $ATET$ s can be heterogeneous over cohorts, across time, across both time and cohorts
- Objective: consistently estimate $ATET$ s and summarize them

Key assumptions

- Observe I.I.D samples of $\{y_{i,t}, \mathbf{x}_{i,t}, \mathbf{z}_{i,t}, d_{i,t}\}_{i=1, t=1}^{i=N, t=T}$, where $\mathbf{x}_{i,t}$ and $\mathbf{z}_{i,t}$ are covariates, and $d_{i,t}$ is observational level treatment indicator
- No one is treated in the first period
- No anticipation in pre-treatment periods $t < g$

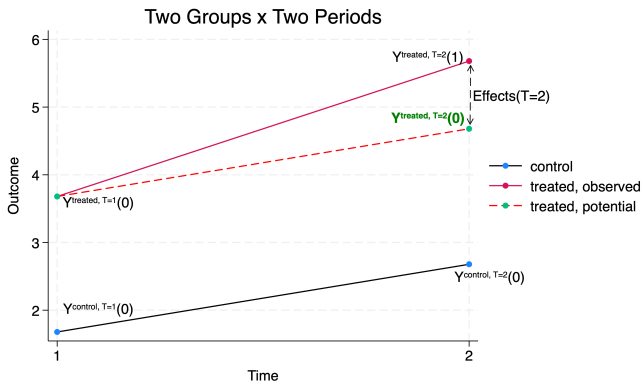
$$\mathbf{E}[y_{i,t}(g)|\mathbf{x}, G_i = g] = \mathbf{E}[y_{i,t}(\infty)|\mathbf{x}, G_i = g]$$

- Conditional parallel trend

$$\mathbf{E}[y_{i,t}(\infty) - y_{i,t-1}(\infty)|\mathbf{x}, G_i = g] = \mathbf{E}[y_{i,t}(\infty) - y_{i,t-1}(\infty)|\mathbf{x}, G_i = \infty]$$

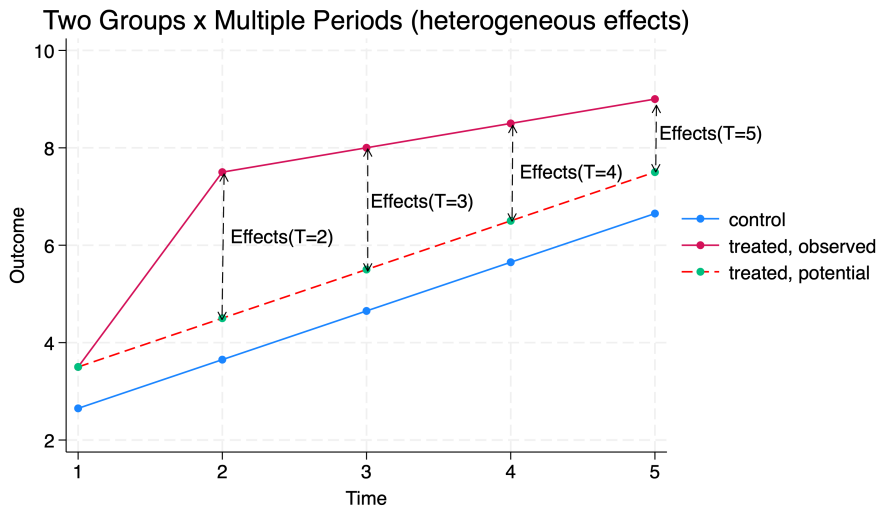
- Overlap assumption for propensity scores

Canonical DID

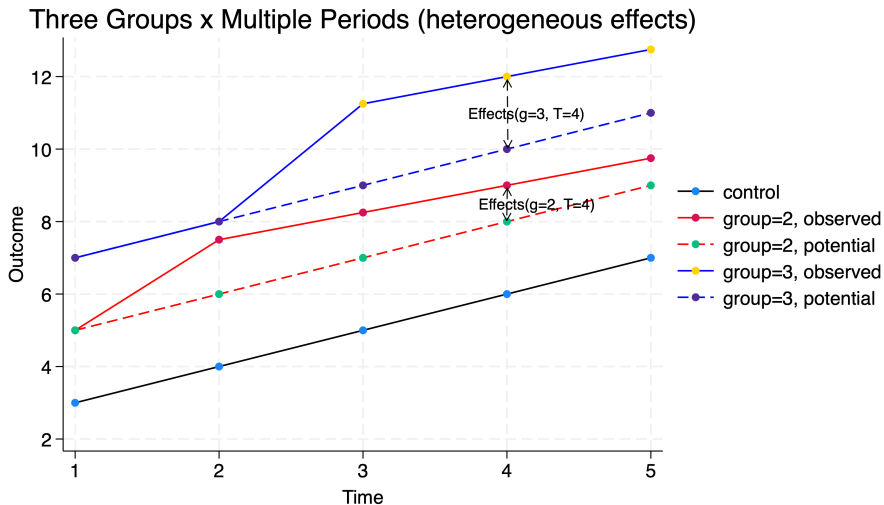


$$\begin{aligned}
 \text{Effects} &= [Y^{\text{treated}, T=2}(1) - Y^{\text{treated}, T=1}(0)] - [Y^{\text{control}, T=2}(0) - Y^{\text{control}, T=1}(0)] \\
 &= \underbrace{[Y^{\text{treated}, T=2} - Y^{\text{treated}, T=1}]}_{\text{treated differences}} - \underbrace{[Y^{\text{control}, T=2} - Y^{\text{control}, T=1}]}_{\text{untreated differences}}
 \end{aligned}$$

Heterogenous DID: benchmark time $g = 1$



$ATET(g, t)$ is reduced to 2×2 DID



Regression adjustment (RA)

$$\begin{aligned} ATET(g, t) &= \mathbf{E} \left[\frac{K_g}{\mathbf{E}(K_g)} (y_t - y_{g-1} - m_{g,t}) \right] \\ &= \underbrace{\mathbf{E} [y_t - y_{g-1} | K_g = 1]}_{\text{treated differences}} - \underbrace{\mathbf{E} [m_{g,t}(\mathbf{x}) | K_g = 1]}_{\text{untreated differences}} \end{aligned}$$

where

- $K_g = \mathbb{1}(G_i = g)$ and $m_{g,t}(\mathbf{x}) = \mathbf{E}(y_t - y_{g-1} | \mathbf{x}, G_i = \infty)$
- It is 2×2 difference-in-differences (two groups $\times$ two periods)
- Benchmark time: one period before treatment ($g - 1$)
- Benchmark group: never-treated group ($G_i = \infty$)

In Stata, we type

- `xthdidregress ra  $\underbrace{(y \quad \mathbf{x})}_{m_{g,t}(\mathbf{x})} (d), \text{group}(id)$`

Inverse probability weighting (IPW)

$$ATE(g, t) = \mathbf{E} \left[\left(\frac{K_g}{\mathbf{E}(K_g)} - \frac{\frac{p_g(\mathbf{z})K_\infty}{1-p_g(\mathbf{z})}}{\mathbf{E} \left[\frac{p_g(\mathbf{z})K_\infty}{1-p_g(\mathbf{z})} \right]} \right) (Y_t - Y_{g-1}) \right]$$

where

- $p_g(\mathbf{z}) = \Pr(K_g = 1 | \mathbf{z}, K_g + K_\infty = 1) = \frac{\Pr(K_g=1|\mathbf{z})}{\Pr(K_g+K_\infty=1|\mathbf{z})}$
- $\frac{p_g(\mathbf{z})}{1-p_g(\mathbf{z})} = \frac{\Pr(K_g=1|\mathbf{z})}{\Pr(K_\infty=1|\mathbf{z})}$. Thus, in the benchmark group (never treated), attach more weights to observations that are more probably observed in the cohort g
- We estimate $p_g(\mathbf{z})$ by a logit regression

In Stata, we type

- `xthdidregress ipw (y)  $\underbrace{(d \ \mathbf{z})}_{p_g(\mathbf{z})}, \text{group(id)}$`

Augmented inverse probability weighting (AIPW)

$$ATE_T(g, t) = \mathbf{E} \left[\underbrace{\left(\frac{K_g}{\mathbf{E}(K_g)} - \frac{\frac{p_g(\mathbf{z})K_\infty}{1-p_g(\mathbf{z})}}{\mathbf{E} \left[\frac{p_g(\mathbf{z})K_\infty}{1-p_g(\mathbf{z})} \right]} \right)}_{\text{IPW}} (Y_t - Y_{g-1} - \overbrace{m_{g,t}(\mathbf{x})}^{\text{augmented term}}) \right]$$

- AIPW is **doubly robust**: only one of the outcome model or the treatment model needs to be correctly specified

In Stata, we type

- `xthdidregress aipw  $\underbrace{(y \ x)}_{m_{g,t}(\mathbf{x})} \underbrace{(d \ \mathbf{z})}_{p_g(\mathbf{z})}, \text{ group(id)}$`

TWFE

Traditional TWFE:

$$y_{it} = \theta_t + \eta_i + d_{it}\alpha + v_{it}$$

TWFE in Wooldridge (2021):

$$y_{it} = \theta_t + \eta_i + \sum_{g \in \mathbf{G}} \sum_{s=g}^T \alpha_{g,t} \mathbb{1}(G_i = g, t = s) + v_{it}$$

With covariates $\mathbf{x}$, add full interactions with θ_t , η_i , and $\mathbb{1}(G_i = g, t = s)$.

In Stata, we type

• `xthdidregress twfe` $\underbrace{(y \ \mathbf{x})}_{\text{TWFE outcome}} \text{ (d) , group(id)}$

Summary

- We illustrate the heterogeneous treatment effects across time and groups
- Traditional TWFE is inconsistent
- `xthdidregress` and `hdidregress` implements four estimators to remedy the heterogeneity issues
- `estat atetplot` visualize the heterogeneity at the group-time level
- `estat aggregation` estimate and visualize the heterogeneity at a higher level

Some extensions

- Switchers: treatments are on and off (de Chaisemartin and D'Haultfœuille, 2020)
- Partial identification: event study when parallel trends assumption is violated (Rambachan and Roth, 2023)
- High-dimensional controls: AIPW estimator is Neyman orthogonal (Callaway et al., 2023)

References

- Callaway, B., D. Drukker, D. Liu, and P. Sant'Anna. 2023. Double/Debiased Machine-learning estimator for Difference-in-Difference with Multiple Periods. URL <https://www.doi.org/10.13140/RG.2.2.33815.65447>.
- Callaway, B., and P. H. Sant'Anna. 2021. Difference-in-Differences with multiple time periods. *Journal of Econometrics* 225: 200–230.
- de Chaisemartin, C., and X. D'Haultfoeulle. 2020. Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects. *American Economic Review* 110: 2964–2996.
- Rambachan, A., and J. Roth. 2023. A more credible approach to parallel trends. *Review of Economic Studies* 90(5): 2555–2591.
- Wooldridge, J. M. 2021. Two-Way Fixed Effects, the Two-Way Mundlak Regression, and Difference-in-Differences Estimators. URL <https://ssrn.com/abstract=3931952>.

Appendix

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1 xthdidregress

```
. global covars i.region pop medinc white hs pov c.pop#c.pop c.medinc#c.medinc
. xthdidregress aipw (lemp $covars) (treat $covars), group(state)
note: variable _did_cohort, containing cohort indicators formed by treatment
      variable treat and group variable state, was added to the dataset.
```

Computing ATET for each cohort and time:

Cohort 2004 (6): done

Cohort 2006 (6): done

Cohort 2007 (6): done

Treatment and time information

Time variable: year

Time interval: 2001 to 2007

Control: _did_cohort = 0

Treatment: _did_cohort > 0

	_did_cohort
Number of cohorts	4
Number of obs	
Never treated	9639
2004	700
2006	1561
2007	4088

Heterogeneous-treatment-effects regression

Number of obs = 15,988

Number of panels = 29

Estimator: Augmented IPW

Panel variable: countyreal

Treatment level: state

Control group: Never treated

(Std. err. adjusted for 29 clusters in state)

Cohort		Robust ATET	std. err.	z	P> z	[95% conf. interval]	
2004	year						
	2002	.0672458	.0061125	11.00	0.000	.0552655	.079226
	2003	.0266718	.0122508	2.18	0.029	.0026608	.0506829
	2004	-.0979371	.002649	-36.97	0.000	-.103129	-.0927451
	2005	-.1139248	.0070092	-16.25	0.000	-.1276627	-.1001869
	2006	-.1719979	.0082852	-20.76	0.000	-.1882366	-.1557592
	2007	-.2078132	.0056814	-36.58	0.000	-.2189485	-.196678
2006	year						
	2002	-.0186685	.0105915	-1.76	0.078	-.0394274	.0020904
	2003	.056737	.0181748	3.12	0.002	.0211151	.0923589
	2004	.0212315	.0363779	0.58	0.559	-.0500679	.092531
	2005	.0319911	.0158191	2.02	0.043	.0009863	.0629959
	2006	-.009851	.0117487	-0.84	0.402	-.0328781	.013176
	2007	-.0510452	.0092241	-5.53	0.000	-.069124	-.0329664

2007							
year							
2002	-.0215125	.014779	-1.46	0.145	-.0504788	.0074538	
2003	.0167167	.0132905	1.26	0.208	-.0093322	.0427655	
2004	.0149363	.0133763	1.12	0.264	-.0112809	.0411534	
2005	.0038453	.0092391	0.42	0.677	-.014263	.0219537	
2006	-.0390546	.0114977	-3.40	0.001	-.0615896	-.0165196	
2007	-.0292338	.0136042	-2.15	0.032	-.0558976	-.00257	

Note: ATET computed using covariates.

2 estat aggregation, dynamic

```
. estat aggreg, dynamic graph(name(d1))
```

Duration of exposure ATET

Number of obs = 15,988

(Std. err. adjusted for 29 clusters in state)

Exposure	ATET	Robust std. err.	z	P> z	[95% conf. interval]	
-5	-.0215125	.014779	-1.46	0.145	-.0504788	.0074538
-4	.0069386	.0100519	0.69	0.490	-.0127627	.0266399
-3	.0264872	.0126915	2.09	0.037	.0016122	.0513621
-2	.0151101	.0118987	1.27	0.204	-.0082109	.0384311
-1	-.0143403	.0124878	-1.15	0.251	-.0388159	.0101353
0	-.032043	.0122219	-2.62	0.009	-.0559975	-.0080885
1	-.0705126	.0161956	-4.35	0.000	-.1022553	-.0387699
2	-.1719979	.0082852	-20.76	0.000	-.1882366	-.1557592
3	-.2078132	.0056814	-36.58	0.000	-.2189485	-.196678

Note: Exposure is the number of periods since the first treatment time.

3 estat aggregation, cohort

```
. estat aggreg, cohort graph(name(c1))
```

ATET over cohort

Number of obs = 15,988

(Std. err. adjusted for 29 clusters in state)

Cohort	ATET	Robust std. err.	z	P> z	[95% conf. interval]	
2004	-.1479183	.0053113	-27.85	0.000	-.1583283	-.1375082
2006	-.0304481	.0075561	-4.03	0.000	-.0452578	-.0156384
2007	-.0292338	.0136042	-2.15	0.032	-.0558976	-.00257

4 estat aggregation, time

```
. estat aggreg, time graph(name(t1))
```

ATET over time

Number of obs = 15,988

(Std. err. adjusted for 29 clusters in state)

Time	Robust		z	P> z	[95% conf. interval]	
	ATET	std. err.				
2004	-.0979371	.002649	-36.97	0.000	-.103129	-.0927451
2005	-.1139248	.0070092	-16.25	0.000	-.1276627	-.1001869
2006	-.0600513	.0406199	-1.48	0.139	-.1396648	.0195622
2007	-.0542855	.0206191	-2.63	0.008	-.0946981	-.0138728